

Professional way to prepare programming contest problems

Statements

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Circle Covers Circle

time limit per test: 2.5 seconds

memory limit per test: 256 megabytes

*[Isabella's Lullaby](#) — a
masterpiece to listen to*
Composer: Takahiro Obata.

At the orphanage in *The Promised Neverland*, Emma, Norman, and Ray are drawing on a graph paper.

Emma and Norman each draw a circle — two perfect shapes on a graph paper. Ray looks at them and decides to draw one more circle that will completely cover both of their drawings.

However, Ray does not want to waste space. He wants his circle to be as small as possible while still covering both circles entirely.

Each circle is described by three integers (x_i, y_i, r_i) , where (x_i, y_i) is the center and r_i is the radius.

Help Ray determine the minimum area of such a circle.

Problem: **circle-circle**, id=526383 
AI tips: Enabled (done)
Owner: Tanvir Ahmed (tanvir_005)
Note:  [click here to add note](#)
Statements: [english](#)
Checker: std::rcmp4.cpp (0)
Validator: [val.cpp](#) (1)
Tests: [tests](#) (27)
Solutions: [gpt.cpp](#) (2/2)
Package: for revision 6
Verification: ok ([restart](#))

[Review problem](#)

Correction tips from AI:

- for the English statement → [review](#)

Access: OWNER, Write:3, Read:1
Revision: 6 / 6
Invokers: 59 / 60

[View changes](#)

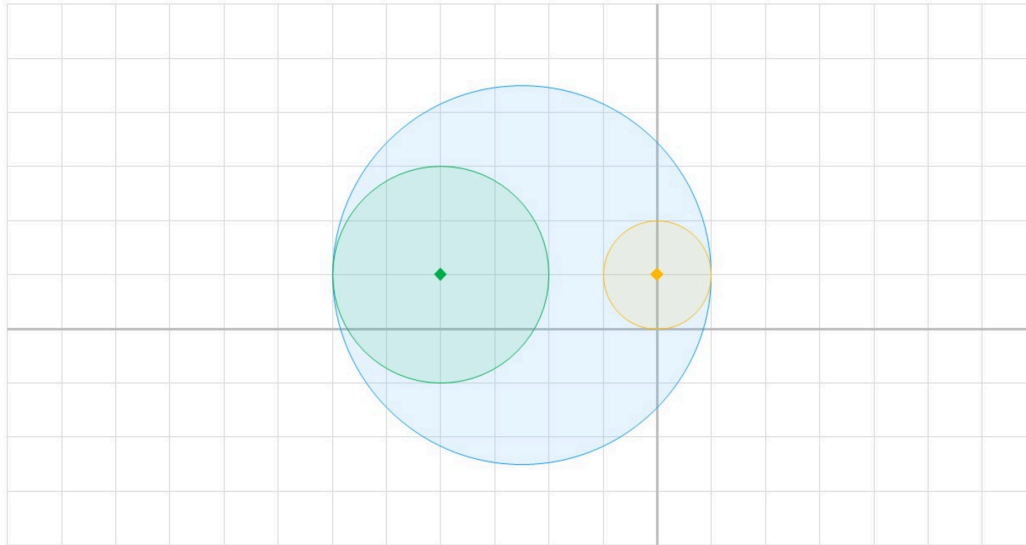
https://polygon.codeforce...anvir_005/circle-circle 

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Emma's Circle: $(-4, 1, 2)$
Norman's Circle: $(0, 1, 1)$



Area of Ray's circle that covers both circles: 38.4845100065 sq units.

Figure: Example representation in graph.

Input

The first line contains a single integer t ($1 \leq t \leq 10^5$) — the number of test cases.

Each test case consists of two lines:

- The first line contains three integers x_1 , y_1 , and r_1 — the center coordinates and radius of the circle drawn by Emma.
- The second line contains three integers x_2 , y_2 , and r_2 — the center coordinates and radius of the circle drawn by Norman.

Additional Constraints: $(-10^5 \leq x_1, x_2, y_1, y_2 \leq 10^5; 1 \leq r_1, r_2 \leq 10^5)$

Output

For each test case, print a single real number — the minimum area of the circle that Ray should draw to completely cover both Emma's and Norman's circles.

Your answer will be accepted if the absolute or relative error does not exceed 10^{-4} .

Example

input	
1 -4 1 2 0 1 1	
output	
38.4845100065	